

M002 Week 4 — AND / OR

Topic: AND and OR Conditions — Smarter Decisions · **Course:** Maze Madness · **Level:** Advanced ·

Time: about 55 minutes

Last week your agent used `and` (both true). This week you add `or` (at least one true) and combine them in one maze.

- `and` → the action runs **only when both** conditions are true.
- `or` → the action runs **when at least one** condition is true.

Detect walls with `agent.detect(BLOCK, FORWARD)`. It returns `True` when a block is there. Use `not` for an open path.

1 · Predict

Read each block. Write what the agent does before you picture it.

```
if agent.detect(BLOCK, FORWARD) and agent.detect(BLOCK, RIGHT):
    agent.turn(LEFT_TURN)
```

Wall ahead, open right. Does it turn left? Why?


```
if not agent.detect(BLOCK, FORWARD) or not agent.detect(BLOCK, DOWN):
    agent.move(FORWARD, 1)
```

Ahead blocked, below open. Does it move? Why?


```
while True:
    if agent.detect(BLOCK, FORWARD) and agent.detect(BLOCK, RIGHT):
        agent.turn(LEFT_TURN)
    elif not agent.detect(BLOCK, FORWARD) or not agent.detect(BLOCK, LEFT):
        agent.move(FORWARD, 1)
    else:
        agent.turn(RIGHT_TURN)
```

At a front-and-right corner, what runs? At an open junction, which branch runs?

2 · Trace the Truth 🔍

Fill the table. Write the branch that runs for each case.

```
if agent.detect(BLOCK, FORWARD) and agent.detect(BLOCK, RIGHT):  
    agent.turn(LEFT_TURN)  
elif not agent.detect(BLOCK, FORWARD):  
    agent.move(FORWARD, 1)  
else:  
    agent.turn(RIGHT_TURN)
```

Front	Right	What runs?
wall	wall	
wall	open	
open	wall	

3 · Spot and Fix Bugs 🐛

Read what each block **should** do, then rewrite it. Explain the fix.

Bug A — Turn left **only** at a dead-end corner (wall ahead AND wall on right). Right now it turns at any wall.

```
if agent.detect(BLOCK, FORWARD) or agent.detect(BLOCK, RIGHT):  
    agent.turn(LEFT_TURN)
```

Fixed code:

Why was it wrong? Why does your fix work?

Bug B — Move forward when **either** the path ahead OR the path below is clear.

```
if not agent.detect(BLOCK, FORWARD) and not agent.detect(BLOCK, DOWN):  
    agent.move(FORWARD, 1)
```

Fixed code:

Why was it wrong? Why does your fix work?

Bug C — At a fork, turn right if the right is open, otherwise turn left. The order is wrong, so it always turns left first.

```
if not agent.detect(BLOCK, LEFT):  
    agent.turn(LEFT_TURN)  
elif not agent.detect(BLOCK, RIGHT):  
    agent.turn(RIGHT_TURN)
```

Hint: check the right side first.

Fixed code:

Why was it wrong? Why does your fix work?

4 · Write the Code

Write a `while True` loop that:

- turns left when there is a wall ahead **and** a wall on the right,
- else moves forward when the path ahead **or** the path below is open,
- else turns right.

5 · Finish the Maze 📷

Switch to your homework world. Solve the maze using **both** an **and** condition and an **or** condition. Come back when the agent reaches the end.

Send a photo or video of the agent at the end. Write 4 to 6 sentences.

First, I ...

I used **and** when ...

I used **or** when ...

One tricky moment was ...

To fix it, I ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Film the maze on your phone. Teach someone who has never seen it. Use these words: **and**, **or**, **condition**, **both**, **either**.

1. Show the start, then run your code.
2. Read each **and** block and **or** block out loud. Say what makes it run.
3. Show one bug you hit and how you fixed it.
4. Say in your own words how **and** is different from **or**.

Plan your words below before you film. You can read from it while recording.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.